

ENGR 213 Final Exam Notes: Applied Ordinary Differential Equations 2023

Premium

Course: Applied Ordinary Differential Equations (ENGR 213) 999+ documents

University: Concordia University

ItzSkyScream

2022/2023

Download

60

0

Save

↑

i v

Concordia University
Faculty of Engineering and Computer Science
Applied Ordinary Differential Equations - ENGR213/4
Winter 2023 Final Examination

Instructors: Korotkin D, Paradis, A., Shnirelman, A.

Special instructions: Do all problems. Each problem worth 10 marks. Show all steps or else no marks will be given. Only Faculty approved calculators are allowed. No other aids are allowed.

1. Find the general solution of the following equation by separation of variables:

$$e^x y \frac{dy}{dx} = e^{-2y} + e^{-3x-2y}$$

2. Find the general solution of the linear differential equation:

$$y dx - 4(x + y^8) dy = 0$$

3. Find the general solution of the homogeneous differential equation:

$$(y^2 + xy)dx - x^2 dy = 0$$

4. Solve the initial-value problem for the following Bernoulli equation:

$$x^2 \frac{dy}{dx} - 2xy = 3y^6 \quad y(1) = 3$$

5. Determine whether the following differential equation is exact:

$$(5x^4y - 15x^2 + 8xy)dx + (x^5 + 4x^2 - 1)dy = 0$$

If yes, find the implicit solution of the initial value problem $y(0) = 1$.

6. Using the method of undetermined coefficients find the general solution of the differential equation

$$y'' - 4y' + 4y = (x - 2)e^{2x}$$

7. Solve the following initial value problem by the method of your choice:

$$x^2 y'' - 4xy' + 6y = 3x^{-3} \quad y(1) = 0, \quad y'(1) = 1$$

Mock exam

Summary

Quiz

Ask a question about this document



8. An LR-series circuit has a variable inductor with the inductance defined by

$$L(t) = \begin{cases} 1 - \frac{t}{10}, & 0 \leq t \leq 10 \\ 0, & t \geq 10 \end{cases}$$

Find the current $i(t)$ if the resistance is 0.2 Ohms, the impressed voltage is $E(t) = 4$ volts and $i(0) = 0$.

9. Find the general solution of the system

$$\frac{dx}{dt} = x - y + 3t$$

$$\frac{dy}{dt} = y - 4x + e^t$$

by the method of your choice.

10. Use the power series method to find the solution of the initial value problem

$$(x-1)y'' + y' + y = 0, \quad y(0) = 1, \quad y'(0) = 2$$

about the point $x = 0$ (write the recursion relation and find first five non-zero terms of the power series solution).